

PERFORMANCE WORK STATEMENT (PWS)

1. SCOPE

The National Oceanic and Atmospheric Administration (NOAA) Global Monitoring Laboratory (GML) Global Greenhouse Gas Reference Network (GGGRN) requires analysis of isotopic ratios, and radio carbon from atmospheric air samples and reference standards from the network.

2. TASKS

A. Isotope Analysis. The vendor shall provide fully quality-controlled isotope analysis of multiple atmospheric trace gases in atmospheric air samples and reference standards from the GGGRN and related programs. The vendor shall take Aliquots from a) pairs of 2.2 liter sample flasks at ~ 1.3 bar in the global network, b) samples within Programmable Flask Packages (PFPs, 0.7 liter flasks at 2.7 bar), and c) reference standard cylinders. Precision requirements are those deemed necessary for scientifically meaningful detection of expected gradients within the atmosphere and are detailed below.

1. Processing and measurement requirements:

- a. Stable isotopes of Carbon Dioxide (CO₂): Provide quantitative isotopic measurement from whole-air atmospheric samples for carbon dioxide including $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$. Results to be reported in per mil (‰) relative to and on scales established by international (IAEA) primary reference materials. The reproducibility of the measurements should be ± 0.02 ‰ and ± 0.05 ‰ for carbon and oxygen respectively. Measurements will be done on 500 cc of air or less. The long term measurement performance must be demonstrated based on past performance in measuring repeat measurements from control cylinders. Quality-control data and meta-data shall be provided upon request.
- b. Radiocarbon (¹⁴C) of Carbon Dioxide (CO₂): Provide i) quantitative, cryogenic extraction and purification of CO₂ from whole-air atmospheric samples, ii) graphitization, and iii) AMS ¹⁴C measurement with iv) combined, long-term (multi-year) external measurement repeatability of 2 ‰ or better (1sigma on $\Delta^{14}\text{C}$ scale). The long-term measurement performance must be demonstrated based on past performance in measuring repeat extraction aliquots of whole air from control cylinders. Quality-control data and meta-data shall be provided upon request.
- c. Radiocarbon (¹⁴C) of Methane (CH₄): Provide i) purification of CH₄ from whole-air atmospheric samples and quantitative conversion to CO₂ for , ii) graphitization, and iii) AMS ¹⁴C measurements with iv) external measurement precision of ≤ 4 ‰ as determined from pair differences or repeat

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measurement control materials. Quality-control data and meta-data shall be provided upon request. Quality-control data and meta-data shall be provided upon request.

- d. Stable isotopes of Methane (CH₄): Provide quantitative isotopic measurement from whole-air atmospheric samples for methane including $\delta^{13}\text{C}$ and δD . Results to be reported in per mil (‰) relative to and on scales established by international (IAEA) primary reference materials. The reproducibility of the measurements should be $\pm 0.08\text{‰}$ and $\pm 2\text{‰}$ for carbon and hydrogen respectively. Measurements will be done on 200 cc of air or less for both isotopes. The long term measurement performance must be demonstrated based on past performance in measuring repeat measurements from control cylinders. Quality-control data and meta-data shall be provided upon request.
- e. Triple oxygen isotope ratios of carbon dioxide ($\Delta^{17}\text{O}$): Provide quantitative isotopic measurement from whole-air atmospheric samples of $\Delta^{17}\text{O}$ of CO₂. Results to be reported in per mil (‰) relative to accepted community scales (none have yet been established by IAEA). The reproducibility of the measurements should be at least $\pm 0.01\text{‰}$. Quality-control data and meta-data shall be provided upon request.

2. Delivery of Sample:

- a. Samples must be transported from GML to the contractor facility. GML will deliver the samples within a five mile radius of the GML facility. Transportation costs beyond the five mile radius is the responsibility of the contractor.
GML Facility Location:
325 Broadway, Boulder, CO 80305
- b. Sample containers shall be returned to NOAA within 50 calendar days or less for individual air samples. GML will retrieve the samples within a five mile radius of the GML facility. Transportation costs beyond the five mile radius is the responsibility of the contractor. Longer delivery times may be allowed with prior authorization from the COR, as instrument downtime dictates, recognizing that these are difficult and not routine measurements. The vendor shall request such delivery times no later than thirty-five (35) days of receipt of sample.

3. Data management and traceability:

- a. The vendor shall have traceability to international reference materials that place sample data on established measurement scales. The vendor shall

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provide, upon request, data that will allow all samples to be post processed at any time to account for subsequent adjustments to reference scales.

- b. The vendor shall provide processed initial results and subsequent quality controlled revisions by digital medium. The vendor shall adjust the format and contents of data files to meet requirements of the NOAA GML database systems as needed.
- c. The vendor shall participate in intercomparison exercises with other labs around the world. This will document the ability of NOAA produced data to be used in conjunction with other data sets without significant scale offsets or measurement artifacts. Coordination will be made with the NOAA GGGRN to make data freely available for this effort.

B. Scientific Interpretation of atmospheric greenhouse gas concentration and isotopic ratio measurements.

1. Radiocarbon (^{14}C) of Carbon Dioxide (CO_2) and Methane (CH_4): The vendor shall provide interpretation of high precision, highly calibrated measurements of the carbon-14 to carbon ratios of atmospheric carbon dioxide and methane made on air samples collected throughout the United States and globally as part of NOAA GML's Global Greenhouse Gas Reference Network. Required activities include:
 - a. Time series analysis of data from individual sites
 - b. Analysis of the north-south gradient of these ratios from multiple sites in the network
 - c. Estimation of emissions and removals of carbon dioxide and methane based on combining isotopic ratio and concentration measurements.
2. Stable isotope ratios of Carbon Dioxide (CO_2) and Methane (CH_4): The vendor shall provide interpretation of high precision, highly calibrated measurements of the carbon-13 to carbon-12 ratios of atmospheric carbon dioxide and methane, and deuterium to hydrogen ratios of methane, made on air samples collected throughout the United States and globally as part of NOAA GML's Global Greenhouse Gas Reference Network. Required activities include:
 - a. Time series analysis of data from individual sites
 - b. Analysis of the north-south gradient of these ratios from multiple sites in the network
 - c. Estimation of emissions and removals of carbon dioxide and methane based on combining isotopic ratio and concentration measurements.

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3. Mole fractions (i.e. “concentrations”) of nitrous oxide (N_2O): The vendor shall provide interpretation of high precision, highly calibrated measurements of atmospheric nitrous oxide made on air samples collected throughout the United States and globally as part of NOAA GML’s Global Greenhouse Gas Reference Network. Required activities include:
 - a. Time series analysis of data from individual sites
 - b. Analysis of the north-south gradient of these ratios from multiple sites in the network
 - c. Estimation of emissions and removals of nitrous oxide based on concentration measurements.

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PERFORMANCE REQUIREMENTS SUMMARY:

Contractor performance will be assessed based on the following criteria.

A. Measurements

Item	Performance Objective (Minimum Acceptable Standard)	Performance Standards	Method of Surveillance	Disincentive
<u>PWS Section</u> <u>2.A.1.a.</u> <u>Stable isotopes of</u> <u>Carbon Dioxide</u> <u>(CO₂)</u>	Provide quantitative measurements from whole-air atmospheric samples for carbon dioxide including $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$. Provide analyses of up to 20,000 samples per year	Results to be reported in per mil (‰) relative to and on scales established by international (IAEA) primary reference materials. The reproducibility of the measurements should be ± 0.02 ‰ and ± 0.05 ‰ for carbon and oxygen respectively. Measurements will be done on 500 cc of air or less. The long term measurement performance must be demonstrated based on past performance in measuring repeat measurements from control cylinders.	COR Inspect and accept	Possible inclusion in performance rating.
<u>PWS Section</u> <u>2.A.1.b.</u> <u>Radiocarbon (¹⁴C)</u> <u>of Carbon Dioxide</u> <u>(CO₂)</u>	Provide for extraction of up to 5,000 authentic samples and measurement of 1,000 to 3,000 samples per year	The long-term measurement performance must be demonstrated based on past performance in measuring repeat extraction aliquots of whole air from control cylinders.	COR Inspect and accept	Possible inclusion in performance rating.
<u>PWS Section</u> <u>2.A.1.c.</u> <u>Radiocarbon (¹⁴C)</u> <u>of Methane CH₄</u>	Provide analyses of up to 400 samples per year	The long-term measurement performance must be demonstrated based on past performance in measuring repeat extraction aliquots of whole air from control cylinders.	COR Inspect and accept	Possible inclusion in performance rating.

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<u>PWS Section 2.A.1.d.</u> <u>Stable isotopes of Methane (CH₄):</u>	Provide quantitative isotopic measurement from whole-air atmospheric samples for methane including $\delta^{13}\text{C}$ and δD. Provide analyses of up to 10,000 samples per year	Results to be reported in per mil (‰) relative to and on scales established by international (IAEA) primary reference materials. The reproducibility of the measurements should be $\pm 0.08\text{‰}$ and $\pm 2\text{‰}$ for carbon and hydrogen respectively. Measurements will be done on a total of 200cc of air or less for both isotopes. The long term measurement performance must be demonstrated based on past performance in measuring repeat measurements from control cylinders.	COR Inspect and accept	Possible inclusion in the performance rate.
<u>PWS Section 2.A.1.e.</u> <u>Triple oxygen isotope ratios of carbon dioxide:</u>	Provide quantitative isotopic measurement from whole-air atmospheric samples. Provide analyses of up to 500 samples per year	Results to be reported in per mil (‰) relative to and on community accepted scales. The reproducibility of the measurements should be $\pm 0.01\text{‰}$. The long term measurement performance must be demonstrated based on past performance in measuring repeat measurements from control cylinders.	COR Inspect and accept	Possible inclusion in the performance rate.

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B. Scientific Interpretation

Item	Performance Objective (Minimum Acceptable Standard)	Performance Standards	Method of Surveillance	Disincentive
<u>PWS Section 2.B.1.</u> <u>Carbon-14 of</u> <u>Carbon Dioxide and</u> <u>Methane</u>	Curve fitting of data from all observation sites using the Thoning et al. ccgerv software package. Model simulation of Northern and Southern Hemisphere mean values. The contractor shall meet with the NOAA technical point of contact to present and discuss measurement data analyses including suggestions for further analyses and uncertain aspects of the current analyses.	The contractor should provide consistent on-time delivery of requested analysis services and clarifications on requests should be made at least one week in advance of meetings.	COR Inspect and accept	Possible inclusion in performance rating.
<u>PWS Section 2.B.2.</u> <u>Stable isotopes of</u> <u>carbon dioxide and</u> <u>methane</u>	Curve fitting of data from all observation sites using the Thoning et al. ccgerv software package. Model simulation of Northern and Southern Hemisphere mean values. The contractor shall meet with the NOAA technical point of contact to present and discuss measurement data analyses including suggestions for further analyses and uncertain aspects of the current analyses.	The contractor should provide consistent on-time delivery of requested analysis services and clarifications on requests should be made at least one week in advance of meetings.	COR Inspect and accept	Possible inclusion in performance rating.
<u>PWS Section 2.B.3.</u> <u>Nitrous Oxide</u>	Inverse modeling of nitrous oxide concentrations to estimate emissions and removals over North America. The contractor shall meet with the NOAA technical point of contact to present and discuss measurement data analyses including suggestions for further analyses and uncertain aspects of the current analyses.	The contractor should provide consistent on-time delivery of requested analysis services and clarifications on requests should be made at least one week in advance of meetings.	COR Inspect and accept	Possible inclusion in performance rating.

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DELIVERABLE SCHEDULE:

A. Measurements

Deliverable	Frequency	# of Copies	Medium/Format	Submit To
Stable isotopes of Carbon Dioxide (CO_2)	Quality-controlled result less than 10 calendar days after quality control.	1	Approved remote access (see 2.A.3.b. above)	Technical POC/COR
Radiocarbon (^{14}C) of Carbon Dioxide (CO_2)	Quality-controlled result less than 10 calendar days after quality control.	1	Approved remote access (see 2.A.3.b. above)	Technical POC/COR
Radiocarbon (^{14}C) of Methane (CH_4)	Quality-controlled result less than 10 calendar days after quality control.	1	Approved remote access (see 2.A.3.b. above)	Technical POC/COR
Stable isotopes of Methane (CH_4)	Quality-controlled result less than 10 calendar days after quality control.	1	Approved remote access (see 2.A.3.b. above)	Technical POC/COR

B. Scientific Interpretation

Deliverable	Frequency	# of Copies	Medium/Format	Submit To
<u>Carbon-14 of Carbon Dioxide and Methane</u>	Delivery of analyses will be ongoing throughout the period of performance and will be scheduled in consultation with the NOAA technical point of contact. Twice-monthly meetings to discuss upcoming and past analysis needs. Graphical and written summaries of requested analyses as requested during meetings. Preliminary analyses should be completed within two weeks of initial request.	1	Remote or in-person meetings and email.	Technical POC/COR
<u>Stable isotopes of carbon dioxide and methane</u>	Delivery of analyses will be ongoing throughout the period of performance and will be scheduled in consultation with the NOAA technical point of contact. Twice-monthly meetings to discuss upcoming and past analysis needs. Graphical and written summaries of requested analyses as requested during meetings. Preliminary analyses should be completed within two weeks of initial request.	1	Remote or in-person meetings and email.	Technical POC/COR
<u>Nitrous Oxide</u>	Delivery of analyses will be ongoing throughout the period of performance and will be scheduled in consultation with the NOAA technical point of contact. Twice-monthly meetings to discuss upcoming and past analysis needs. Graphical and written summaries of requested analyses as requested during meetings. Preliminary analyses should be completed within two weeks of initial request.	1	Remote or in-person meetings and email.	Technical POC/COR